Complexity-Aware Method Selection: A Three-Step Diagnostic Workflow

Diagnose the model–data complexity profile, then select an XAI method whose assumptions match it (§2.4)

STEP 1 — Diagnose the complexity profile

Run five quantitative tests on the trained model + data (cheapest first). Each flags one dimension; all cutoffs are provisional (Table 2).

Collinearity	Interactions	Nonlinearity	Suppressors	Capacity
VIF, condition no. $\kappa(X)$	SHAP interaction matrix	degree-2 poly vs. linear ΔR^2	r vs. partial corr ρ	eff. Hessian rank
flag: VIF > 5 (moderate), > 10	flag: $\lvert\Phi jk\rvert > 0.05 \times$ mean main	flag: NL > 0.10	flag: sign flip or $\lvert\rho\rvert > 2\lvert r\rvert$	flag: leaves > 10^5 / rank > 100

Any test flagged?

NO

YES

Complexity-simple

Standard SHAP / LIME / IG are likely reliable.

STEP 2 — Map diagnosis to method

If multiple flags, provisional tie-breaker (not a severity ranking):

interactions > suppressors > collinearity > nonlinearity > capacity

- Interactions → TreeSHAP interaction values, Shapley-Taylor, Integrated Hessians
- Suppressors → PatternAttribution / Haufe patterns; report the suppressor check
- Collinearity → grouped interv. SHAP (model); obs. SHAP, CALIME, AEC (data)
- Nonlinearity → IG, Integrated Hessians, DeepLIFT
- Capacity → SmoothGrad, VarGrad, ensemble attributions
- Architecture → SLALOM (transformers); applicability: failure-mode map (§3)

STEP 3 — Validate the diagnosis

1. Apply one recommended and one discouraged method — rankings should diverge on flagged features.
2. Model questions: ROAR — recommended method should give a steeper drop (confounded by redundancy).
3. Stability: perturb input by ε; recommended method's attributions should change less.
4. Agreement ($\tau > 0.9$) despite a flag → false positive or no effect (interactions: compare ISD readouts).

Outcome: an evidence-based method choice instead of a default. Worked end-to-end in §2.5; illustrated in E1–E10 and on real data in E4c (§5).